

Mobius Breez Microbioreactor - 2.155 mL working volume

merck-breez

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	2.155 mL	Derived / estimated
Maximum working volume	2.155 mL	Source-supported
Nominal vessel / bag size	2 mL	Source-supported
Total vessel volume loaded into the form	2.155 mL	Derived / estimated
Initial working volume loaded into the form	2.155 mL	Derived / estimated
Product-contact material	Plastic / single-use	App assumption
Height : diameter	2	App assumption
Internal / chamber diameter	0.0111 m	Derived / estimated
Liquid height at maximum working volume	0.0222 m	Derived / estimated
Impeller / mixing type	Marine / placeholder mixing element	App assumption
Impeller count	1	App assumption
Impeller diameter	0.003667 m	Derived / estimated
Power number	0.1	App assumption
Initial agitation	100 rpm	App assumption
Maximum agitation	500 rpm	App assumption
Initial aeration	0.05 vvm	App assumption
Maximum aeration	0.2 vvm	App assumption
Baffled vessel	No	App assumption
Maximum pressure	0.2 bar(g)	App assumption
Default operating pressure	0 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 2.155 mL

Derived / estimated | Merck Mobius Breez datasheet DS12130EN (online only) - PDF page 6; Specifications - Fluid Flow and Volume

Rule: minimum = specified fixed working volume

Selection / calculation: $2.155 \text{ mL} / 1000 = 0.002155 \text{ L}$

Limitations: The manufacturer specifies a fixed working volume, not a separate minimum. The app sets both endpoints equal for a fixed-volume preset.

Original unavailable locally; primary evidence observed online.

[Manufacturer publication](#)

Maximum working volume - 2.155 mL

Source-supported | Merck Mobius Breez datasheet DS12130EN (online only) - PDF page 6; Specifications - Fluid Flow and Volume

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2.155 mL.

Limitations: Exact specified fixed working volume, more precise than the 2 mL marketing description. Read online 2026-09-11; local original download blocked.

Original unavailable locally; primary evidence observed online.

[Manufacturer publication](#)

Nominal vessel / bag size - 2 mL

Source-supported | Merck Mobius Breez datasheet DS12130EN (online only) - PDF page 1; Product description

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2 mL.

Limitations: Named 2 mL platform, distinguished from the exact 2.155 mL working volume. Total capacity remains unverified.

Original unavailable locally; primary evidence observed online.

[Manufacturer publication](#)

Total vessel volume loaded into the form - 2.155 mL

Derived / estimated | Form-population rule

Rule: total volume = $\max(\text{nominal vessel volume}, \text{maximum working volume})$

Selection / calculation: $\max(2 \text{ mL}, 2.155 \text{ mL}) = 2.155 \text{ mL}$

Limitations: This prevents the form's total vessel volume from being smaller than the working-volume ceiling.

Initial working volume loaded into the form - 2.155 mL

Derived / estimated | Form-population rule

Rule: initial working volume = $\min(\text{minimum working volume}, \text{maximum working volume})$

Selection / calculation: $\min(2.155 \text{ mL}, 2.155 \text{ mL}) = 2.155 \text{ mL}$

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Plastic / single-use

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Plastic / single-use.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Height : diameter - 2

App assumption | Generic microfluidic profile

Rule: PROFILE_DEFAULTS["microfluidic"].heightDiameterRatio = 2

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microfluidic starting profile. This ratio is subsequently used to estimate vessel diameter and liquid height. This is a microfluidic system. These stirred-tank fields are compatibility placeholders for the current simulation engine and should not be interpreted as physical impeller specifications.

Internal / chamber diameter - 0.0111 m

Derived / estimated | Cylindrical geometry estimate

Rule: $D = \sqrt[3]{(4 \times V) / (\pi \times H : D)}$

Selection / calculation: $D = \sqrt[3]{(4 \times 0.00002155 \text{ m}^3) / (\pi \times 2)} = 0.01111158 \text{ m}$; the form displays 0.0111 m.

Limitations: The estimate treats the liquid volume as a straight-sided cylinder and does not include heads, cones, internals or manufacturer-specific dimensions.

Liquid height at maximum working volume - 0.0222 m

Derived / estimated | Cylindrical geometry estimate

Rule: liquid height = estimated internal diameter x H:D

Selection / calculation: $0.01111158 \text{ m} \times 2 = 0.02222316 \text{ m}$; the form displays 0.0222 m.

Limitations: This is linked to the same ideal-cylinder approximation used for the diameter.

Impeller / mixing type - Marine / placeholder mixing element

App assumption | Generic microfluidic profile

Rule: PROFILE_DEFAULTS["microfluidic"].impellerType = "marine"

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microfluidic starting profile. The selected type determines the generic mixing archetype shown and is paired with the preset power-number assumption. This is a microfluidic system. These stirred-tank fields are compatibility placeholders for the current simulation engine and should not be interpreted as physical impeller specifications.

Impeller count - 1

App assumption | Generic microfluidic profile

Rule: PROFILE_DEFAULTS["microfluidic"].impellerCount = 1

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microfluidic starting profile. The count is used by the vessel diagram and represents a generic configuration unless source-supported. This is a microfluidic system. These stirred-tank fields are compatibility placeholders for the current simulation engine and should not be interpreted as physical impeller specifications.

Impeller diameter - 0.003667 m

Derived / estimated | Generic impeller-to-tank ratio

Rule: impeller diameter = $\max(0.33 \times \text{estimated vessel diameter}, 0.001 \text{ m})$

Selection / calculation: $\max(0.33 \times 0.01111158 \text{ m}, 0.001 \text{ m}) = 0.003666821 \text{ m}$; the form displays 0.003667 m.

Limitations: A one-third tank-diameter impeller is used only as a generic starting geometry when no model-specific impeller diameter is stored.

Power number - 0.1

App assumption | Generic microfluidic profile

Rule: PROFILE_DEFAULTS["microfluidic"].powerNumber = 0.1

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microfluidic starting profile. The power number is an empirical generic value used in $P = N_p N_3 D_i^5$. It is not recalculated from blade geometry. This is a microfluidic system. These stirred-tank fields are compatibility placeholders for the current simulation engine and should not be interpreted as physical impeller specifications.

Initial agitation - 100 rpm

App assumption | Generic microfluidic profile

Rule: PROFILE_DEFAULTS["microfluidic"].baseRpm = 100

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microfluidic starting profile. This is the initial agitation setting loaded into the form. This is a microfluidic system. These stirred-tank fields are compatibility placeholders for the current simulation engine and should not be interpreted as physical impeller specifications.

Maximum agitation - 500 rpm

App assumption | Generic microfluidic profile

Rule: PROFILE_DEFAULTS["microfluidic"].maxRpm = 500

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microfluidic starting profile. This is the ceiling available to the generic DO-control cascade. This is a microfluidic system. These stirred-tank fields are compatibility placeholders for the current simulation engine and should not be interpreted as physical impeller specifications.

Initial aeration - 0.05 vvm

App assumption | Generic microfluidic profile

Rule: PROFILE_DEFAULTS["microfluidic"].baseVvm = 0.05

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microfluidic starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 0.2 vvm

App assumption | Generic microfluidic profile

Rule: PROFILE_DEFAULTS["microfluidic"].maxVvm = 0.2

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microfluidic starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - No

App assumption | Generic microfluidic profile

Rule: PROFILE_DEFAULTS["microfluidic"].baffled = false

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microfluidic starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 0.2 bar(g)

App assumption | Generic microfluidic profile

Rule: PROFILE_DEFAULTS["microfluidic"].maxPressure = 0.2

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microfluidic starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0 bar(g)

App assumption | Generic microfluidic profile

Rule: PROFILE_DEFAULTS["microfluidic"].operatingPressure = 0

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microfluidic starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

This is a microfluidic perfusion system, not a conventional stirred tank. Mixing-related fields are model placeholders.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.